UNIT 2 THE MOTORCYCLE

Structure

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2.1 INTRODUCTION

In Unit 1, you studied about the growth of service industry, automobile industry, two-wheeler industry and the growth of Hero Honda Motors Ltd. Now you will study different parts, systems and mechanisms of Motorcycle. The Starter Kit would familiarise a technician with basic concepts of a motorcycle. It is important to understand the basic functions of the motorcycles to get a complete picture as to how does a motorcycle work.

Objectives

After studying this unit, you should be able to

- describe how does a motorcycle work,
- identify the transmission system and its various part,
- explain about the fuel feed system,
- appreciate the importance of the suspension system,
- describe the necessity of brakes and tyres, and
- explain the different grades of lubricants.

2.2 GENERAL DESCRIPTION OF A MOTORCYCLE

2.2.1 Basic Concept

Like any fine machine a motorcycle can be split into various systems like lubrication system, transmission system, electrical system, etc.

It is important that a technician understands the basics of a motorcycle and its various components so that he can understand the functioning and importance of each and every part. This would help him to do quality service and ensure that the troubleshooting is done according to the standards set by the company.

It may happen that a particular problem is related to manufacturing of more than one components of the motorcycle. For example, if there is misfiring in the motorcycle it can be due to faults in one or more of the following :

- Spark Plug
- Spark Plug Adaptor
- CDI Unit
- Exciter Coil
- High tension (HT) Coil Primary
- High tension (HT) Coil Secondary
- Pulser Coil
- Ignition Coil
- Ignition Switch
- Others

We shall discuss about these later. It is essential to understand all the basic functions of a motorcycle and its parts that perform these prescribed functions.

2.2.2 Parts of Motorcycle

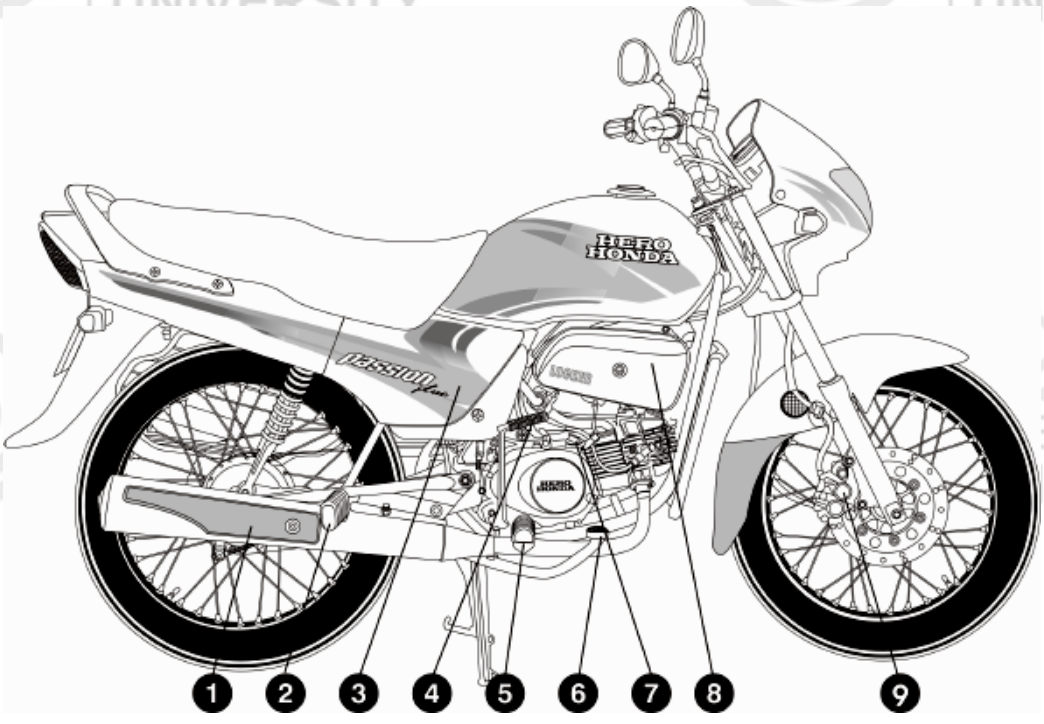


Figure 2.1 : Right Side View of a Motorcycle

Part No.	Name	Functions
1	Muffler heat protector	Protection from hot silencer
2	Pillion foot rest	For the foot rest of pillion rider
3	Battery compartment	To place battery
4	Kick starter pedal	Used to crank engine
5	Rider foot rest	Foot rest for rider/driver
6	Brake pedal	Used to apply brakes
7	Oil filter cap/dipstick	Used to check engine oil level
8	Locker	Compartment to keep tool kit and accessories
9	Caliper assembly	Braking of vehicle

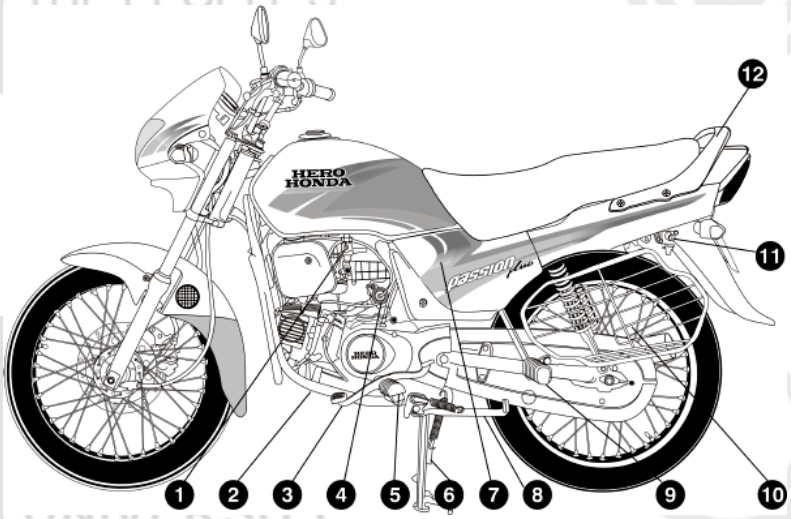


Figure 2.2 : Left Side View of a Motorcycle

Part No.	Name	Functions
1	Fuel cock	Used to open or close the fuel supply
2	Engine guard	For the safety of engine
3	Gear shift pedal	Used to change the gears
4	Air suction valve	Used to supply air to exhaust manifold
5	Rider foot rest	Foot rest for rider/driver
6	Main stand	Used to mount the bike in vertical position
7	Left Side cover	Cover the body parts
8	Side stand	Used to mount the bike in a tilted position
9	Pillion foot rest	For the foot rest of pillion
10	Sari guard	Guard to protect sari from coming into wheel
11	Seat cum helmet lock	To lock seat and helmet
12	Rear grip	Support and grip for pillion rider

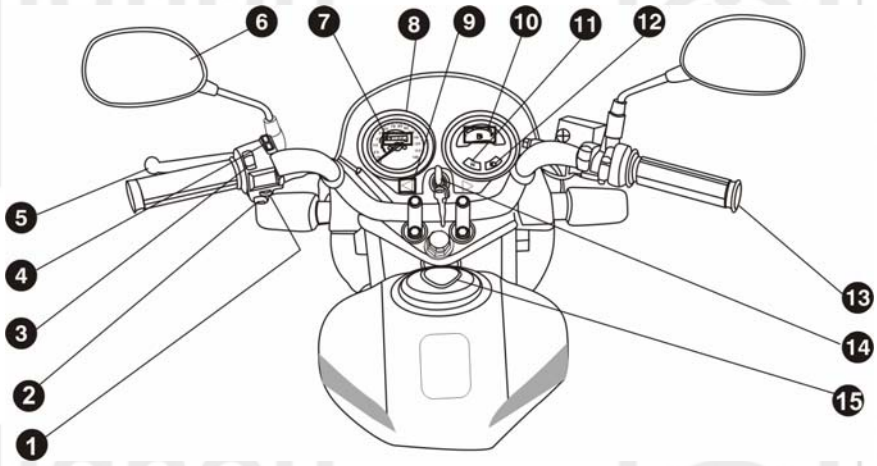


Figure 2.3 : Front Top View of the Motorcycle

Part No.	Part Name	Functions
1	Turn signal switch	Used to give an indication while turning
2	Horn switch	Used to operate the horn
3	Headlight dimmer switch	Used to operate low beam of Headlight
4	Headlight switch	Used to operate headlight
5	Clutch lever	Used to operate clutch (to dis-engage engine power)
6	Rear view mirror	Used to view vehicles coming from the rear
7	Odometer	Shows accumulated mileage
8	Speedometer	Indicates driving speed
9	Indicator lights	Used to show that this vehicle is going to turn in indicated direction
10	Fuel indicator	Used to show the approximate fuel quantity in fuel tank
11	Neutral lights	Used to show the neutral position of gear
12	High beam indicator	Used to show the high beam operation of headlight beam
13	Throttle grip	Used to control throttle valve
14	Ignition switch cum steering lock	Used to start as well as to lock the bike
15	Fuel tank cap	Used to close fuel tank

2.2.3 Two Wheeler Engines

Engine is a unit of motorcycle that runs the motorcycle by converting fuel energy to mechanical power. This power comes from the burning of fuel inside the cylinder of the engine.

The engine of a motorcycle can be classified as follows :

(a) According to Engine Strokes

(i) **Two-stroke Engine :** One power stroke in one revolution of engine crankshaft.

(ii) **Four-stroke Engine :** One power stroke in two revolution of engine crankshaft.

(b) According to Cooling of engine :

(i) **Liquid Cooled Engine**

(ii) **Air Cooled Engine :** These can be of two types as given below :

- Natural Air Cooled Engine
- Forced Air Cooled Engine

Here, we shall study four-stroke petrol engine only because this is used in Hero Honda motorcycles. The petrol engines are also known as spark ignition (SI) engines.

Working of a Four-stroke Engine

To understand the working of a four-stroke motorcycle engine, the whole operation can be divided into four parts, known as strokes. The strokes are named after their functions such as, Suction stroke, Compression stroke, Power or Expansion stroke and Exhaust stroke. Hereunder, in Figure 2.4, the line diagram is showing all the four strokes of operation of an engine.

Figure 2.4 : Engine Diagram

Note : The direction of arrows in Figure 2.4 indicates the direction of movement of piston.

Suction Stroke

The pressure decreases from atmospheric to some lower value inside the cylinder. This happens because of downward piston movement. Because of the pressure difference of outside and inside the cylinder, the charge, which is mixture of air and fuel vapour, is sucked into the cylinder.

Compression Stroke

The charge is compressed during this stroke. The compression enhances the fuel air mixing and makes it closer to perfect mixture, and then a sparking takes place at the end of compression stroke.

Power Stroke

In the beginning of this stroke, the spark causes combustion and the gases expand, exerting pressure on the piston. By the virtue of expansion of gases the pressure increases inside the cylinder on the top of the piston. Because of pressure exerted on the piston, piston moves downward and rotates the crankshaft via connecting rod.

Exhaust Stroke

At the end of the power stroke the exhaust valve opens. The piston moves upward and pushes the burnt gases out.

Even though reciprocating internal combustion engines look quite simple, yet these are highly complex machines. These have hundreds of components, which have to perform their functions satisfactorily to produce power. Before going through the working principle of this complex machine, a brief description of the important engine components and the nomenclature is given below :

Basic Components of Engine

The major components of the engine and their functions are briefly given hereunder. Figure 2.5 shows a cut model of motorcycle engine, where various internal parts are visible.

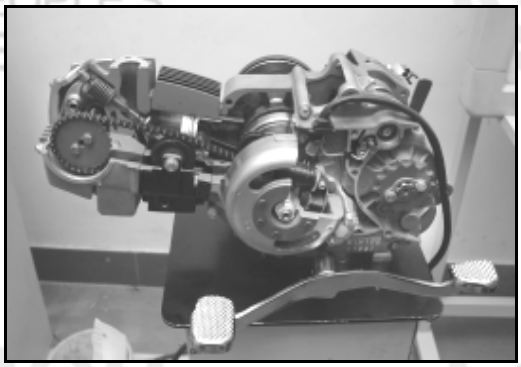


Figure 2.5 : Basic Components of Engine

Cylinder Block

This serves as an enclosure to the cylinder. The cylinder is an integral part of cylinder block in 100 cc motorcycle (Figure 2.6).

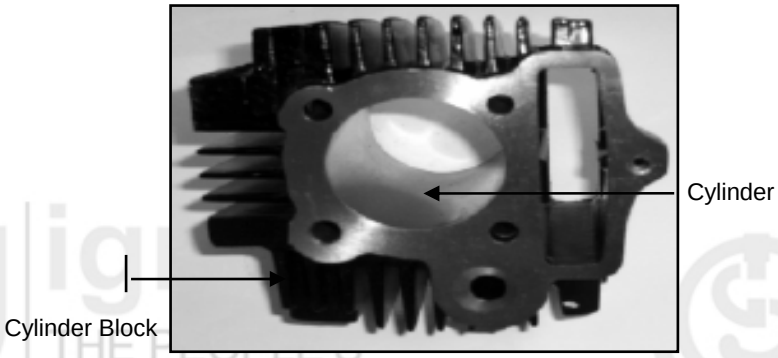


Figure 2.6 : Cylinder Block

Cylinder

As the name implies it is a cylindrical vessel or space in which the piston makes a reciprocating motion (forward and backward). The varying volume created in the cylinder during the operation of the engine is filled with the working fluid (air fuel mixture) subjected to different processes such as suction, compression, expansion and exhaust.

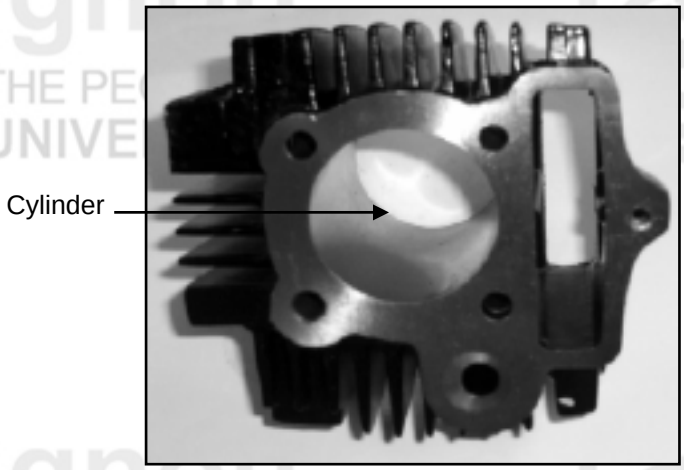


Figure 2.7 : Cylinder

Crankcase

It is connected to engine cylinder. It splits into two parts, named as right crankcase and left crank case. It supports the kick-starter spindle, crankshaft, main shaft and counter shaft. The right and left halves of crank case are shown in Figures 2.8 and 2.9, respectively.

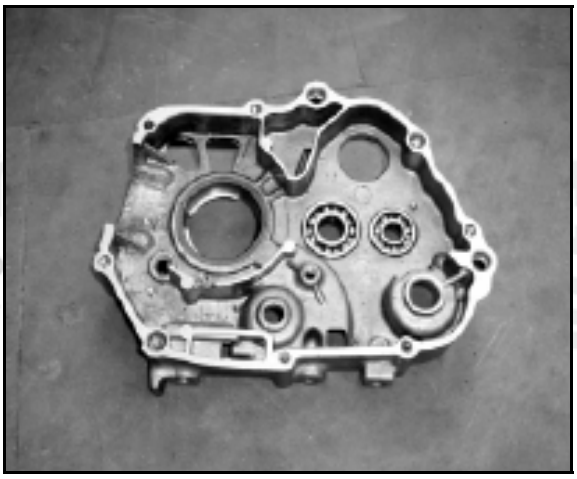


Figure 2.8 : Right Crankcase



Figure 2.9 : Left Crankcase

Cylinder Head

It is fitted at top of the cylinder block. It provides space for the arrangement of spark plug, combustion chamber, camshaft, cam chain, valves and intake and exhaust manifolds.

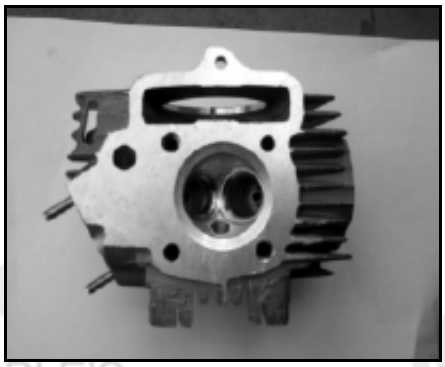


Figure 2.10 : Cylinder Head

Combustion Chamber (*or Clearance Volume*)

Combustion chamber is a space between the cylinder head and the piston top when the piston is at top dead center (TDC). The combustion of air fuel mixture and consequent release of heat results in the building up of pressure in this part of the cylinder.

Piston

It is a cylindrical component fitted into the cylinder forming the moving boundary of the combustion region (Figure 2.11). It fits perfectly (snugly) into the cylinder providing a gas-tight space with the help of piston rings and the lubricant. It forms the first link in transmitting the force exerted by combustion gases to the crankshaft.



Figure 2.11 : Piston

Functions of the Piston

- To transmit force of explosion to the crankshaft.
- To form a seal so that high pressure gases from the combustion chamber do not escape into the crankcase so as to cause loss of power or spoiling of engine oil.
- To serve as a guide and a bearing for small end of the connecting rod.

Desirable Characteristics of the Piston

- Silent in operation.
- Seizure between piston surface and inside of cylinder should not occur.
- Sufficiently resistant to corrosion against combustion.
- Short in length so that engine size (length of cylinder) is decreased.
- Light in weight.
- High thermal conductivity to help quick heat transfer to reduce temperature of top of the piston (for higher compression ratio and to avoid detonation which may otherwise be caused due to pre-ignition).

The piston is subjected to side thrust when the connecting rod is inclined with respect to the cylinder axis. The skirt is provided with bosses in the inside to support piston pin. The skirt must be of sufficient length to resist tilting of the piston under load. It is kept quite close fitted in the cylinder but even then it is separated from the cylinder walls by means of a thin film of lubricating oil. This oil film helps in smooth running of engine.

The method of eliminating slapping tendency of pistons is to offset them from the cylinder axis (center line) towards the major thrust side. The piston pin-hole is not symmetrically placed as usual, but is slightly offset.

As the piston approaches TDC during compression stroke, the offset causes it to tilt slightly so that its top skirt surface on the minor thrust side is placed against cylinder wall.

Simultaneously the bottom surface of the piston skirt on the major thrust side is also placed against the cylinder wall. When the TDC is passed the inclination of the connecting rod forces the piston against the major thrust surface. However, since the lower surface of the skirt is already in contact with the cylinder wall because of the offset the piston skirt very smoothly wipes into full wall contact, thus avoiding piston slap.

If the piston offset is located towards the minor thrust side, a higher mechanical advantage is obtained. Moreover, lesser piston to cylinder friction is caused. However, it will not decrease the piston slap. Therefore in racing car engines the pistons are offset towards minor thrust side since noise and life are not as important as performance in such engines.

Piston Rings

Piston rings (Figure 2.12), fitted into the grooves around the piston, provide a tight sealing effect between the piston and the cylinder wall thus preventing leakage of combustion gases or loss of compression during the compression stroke.

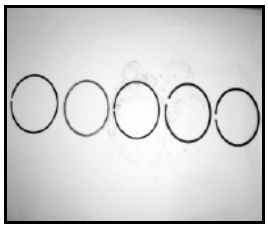


Figure 2.12 : Piston Rings

Functions of the Piston Ring

- To form a seal for high pressure gases from the combustion chamber, against leak into the crankcase.
- To provide sufficient lubricating oil on cylinder walls throughout the entire length of the piston travel, minimizing the ring and cylinder wear, and at the same time control the thickness of the oil film so that satisfactory oil film is maintained. The oil is not to be allowed to go up to the combustion chamber where eventually it would burn to leave carbon deposits on cylinder wall, cylinder head, piston or at the exhaust manifold and on valves seating (or piston pin).

Gudgeon Pin

It forms the link between the small end of the connecting rod and the piston. It is a small pin as shown in Figure 2.13.

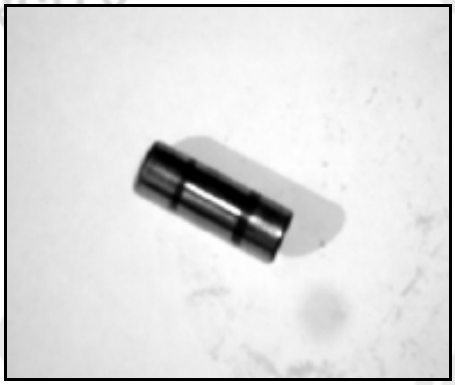


Figure 2.13 : Gudgeon Pin

Inlet Manifold

The pipe which connects the intake system to the inlet valve of the engine through which air or air-fuel mixture is drawn into the cylinder is called the inlet manifold.

Exhaust Manifold

The pipe which connects the exhaust system to the exhaust valve of the engine through which the products of combustion escape into the atmosphere is called the exhaust manifold.

Inlet and Exhaust Valves

Valves are commonly mushroom shaped poppet type. They are provided on the cylinder for regulation of the charge coming into the cylinder (inlet valve) and for discharging the products of combustion (exhaust valve) from the cylinder.

Connecting Rod

It connects the piston and the crankshaft and transmits the power from the piston to the crankshaft. It has a small end (through which gudgeon pin passes to connect with piston) and a bigger end through which it is connected to crankshaft. It is shown in Figure 2.14.



Figure 2.14 : Connecting Rod

Crankshaft

It converts the reciprocating motion of the piston into useful rotary motion. In the crankshaft of a single cylinder engine, there is a pair of crank arms and balance weights. The balance weights (counter weight) are provided on the crankshaft. It balances the weight of various components of rotating system. The crankshaft (Figure 2.15) is enclosed in the crankcase.

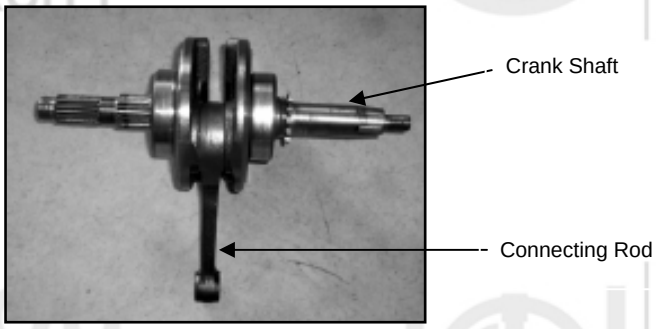


Figure 2.15 : Crankshaft

Camshaft

The camshaft controls the opening and closing of the two valves. The associated parts are push rods, rocker arms and valve springs. The camshaft is driven by a camchain.

Cam

This is an integral part of camshaft. A cam is designed in such a way as to open or close the valves at the correct timing and keep them open for the necessary duration.

Engine Terminology

Figure 2.16 illustrates the nomenclature of an engine :

Figure 2.16 : Nomenclature of Engine

Cylinder Bore

The nominal inner diameter of the working cylinder is called the cylinder bore and is designated by the letter *D* and is usually expressed in millimetre (mm).

Piston Area

The area of a circle of diameter equal to the cylinder bore is called the piston area and is designated by the letter A and is usually expressed in square centimetre (cm^2).

Stroke

The nominal distance through which a working piston moves between two successive reversals of its direction of motion is called the stroke and is designated by the letter *L* and expressed usually in millimetre (mm).

Dead Centre

The position of the working piston and the moving parts which are mechanically connected to it, at the moment when the direction of the piston motion is reserved at either end of the stroke is called the dead centre. There are two dead centers in the engine. They are Top Dead Centre and Bottom Dead Centre.

Figure 2.17 : Dead Centres

Top Dead Centre (TDC) : It is the dead centre when the piston is farthest from the crankshaft. It is designated as TDC. It is also called the Inner Dead Centre (IDC).

Bottom Dead Centre (BDC) : It is the dead centre when the piston is nearest to the crankshaft. It is designated as BDC. It is also called the Outer Dead Centre (ODC).

Displacement or Swept Volume (V_s)

The nominal volume swept by the working piston when travelling from one dead centre to the other is called the displacement volume. It is expressed in terms of cubic centimetre (*cc*)

$$V_s = A \times L$$

Clearance Volume (V_C)

The nominal volume of the combustion chamber above the piston when it is at the top dead centre is the clearance volume. It is designated as V_C and expressed in cubic centimeter (*cc*).

Figure 2.18 : Displacement Volume and Clearance Volume

Compression Ratio (r)

It is the ratio of the total cylinder volume when the piston is at the bottom dead centre, V_t to the clearance volume, V_C.

$$r = \frac{V_t}{V_c}$$

SAQ 1

(a) Classify the two wheeler engines.

(b) Define the following components and their uses in engine with the help of engine diagram.

(i) Crankcase

(ii) Piston and Piston pin

(iii) Connecting rod

(iv) Crankshaft and cam

(v) Inlet and Exhaust valve

2.2.4 Lubrication System

This is an arrangement that consists of oil pump, oil sump, oil lines and lubricant. The lubricant is meant to lubricate the engine moving parts and it also cleans the moving parts of the engine.

Necessity of Lubrication

- To lubricate the various parts of engine.
- The supplementary function of lubrication is to cool and clean the engine parts.
- To prevent engine parts from rusting.
- To provide sealing for combustion gases from entering into sump.
- To damp down engine noise.

Types of Lubricant

Lubricant is a material which circulates through the lubrication network to provide lubrication at various points. Lubricants are classified in three forms: Fluid oils, semi solid and solids.

Fluid Oil

These are liquid oils and used in engine lubrication system.

Semi Solids

Such as Grease. These types of lubricant are used in chassis lubrication system.

Solid

Such as graphite and mica. Such types of lubricant are used in journal bearing and springs with some fluid oils.

Fluid oils can also be classified according to their sources – as animal, vegetable, and mineral oil. Mineral oils are used for lubricating automotive engines because they have most of the desired qualities and simultaneously they are available easily and cheap. The function of a lubricant is to decrease friction by preventing direct contact of rubbing surfaces. It makes a layer between adjacent surfaces. The ability to keep two surfaces apart depends on its various properties.

Various Properties of Lubricant

Viscosity

The property of lubricant which is a measurement of the internal friction of lubricating oil is called its viscosity. The viscosity of an oil is specified in time (seconds) taken by the given quantity of oil to flow freely through a standard size hole at given temperature.

Viscosity Index

Every lubricant thins out when its temperature increases and gets heavier as temperature decreases but this change does not occur at the same rate for every lubricant. A number is given to indicate the rate at which it thins out or gets heavier; this is known as viscosity index.

API Classification

American Petroleum Institute (API) gives a classification for various oils as per the ingredients in the oil. Some of the functions of these ingredients are : anti-foaming, anti-septic, anti-rust, etc.

The classification starts from

SA, SB, SC, . . . , SJ . . . etc.

SAE Number

SAE viscosity numbers are recommended by Society of Automotive Engineers (SAE). The range of viscosities within which it falls at the given temperature determines the SAE viscosity number of any lubricating oil. The lubricating oil recommended by Hero Honda Motors Ltd for their 100 cc models is of grade range SAE 20 W 40 SJ and for 125 cc is SAE 10 W 30 SJ grade.

The lubricating oils of the following SAE grade range are generally available in market.

Multi Viscosity Graded Motor Oils

SAE Grade Range	10W-20	20W-30	20W-40	W-30	5W-20

Type of Lubrication Systems

The complete lubrication of a vehicle can be divided into two systems.

(a) Engine lubrication system.

(b) Chassis lubrication system.

Engine Lubrication System

This system facilitates the lubrication of all the engine components, gears and other associated parts. The brief description of engine lubrication system is given below.

Working of Engine Lubrication System

The oil flows from oil pump into the distribution chamber, where it is divided into two parts. One part enters into the right crankcase cover and flows into the centrifugal rotor. Due to centrifugal action the impurities are collected at the periphery and clean oil goes inside the crankshaft to the crankpin. Through the opening on either side of the connecting rod at crank pin area, the oil flows up and lubricates piston, piston rings, piston pin and cylinder liner.

The second part of oil goes through one of studs to the cylinder head, hence it goes into the camshaft and comes out of small holes on the cam lobe area. This oil lubricates valve, spring, rocker arm etc. The oil also comes out from dowel pin, which has cam sprocket installed on to it. This oil lubricates the cam chain.

Figures 2.19(a) and (b) show engine lubricating system of 100 cc and 125 cc of engines.

The working of the lubrication system of 125 cc is little bit different from 100 cc Motorcycle. An extra oil passage is provided which goes downside from oil pump and lubricates piston skirt and its nearby area. The second difference is the lubrication of gears on main and counter shaft. An extra passage for lubrication oil is provided to lubricate transmission gears.

Figure 2.19(a) : Engine Lubrication System of a 100 CC Engine of HHML Motorcycle

Figure 2.19(b) : Engine Lubrication System of a 125 CC Engine of HHML Motorcycle

Chassis Lubrication System

This lubrication system consists of outer body parts, which are lubricated from outside of the body. The points for lubrication are shown in Figure 2.20.

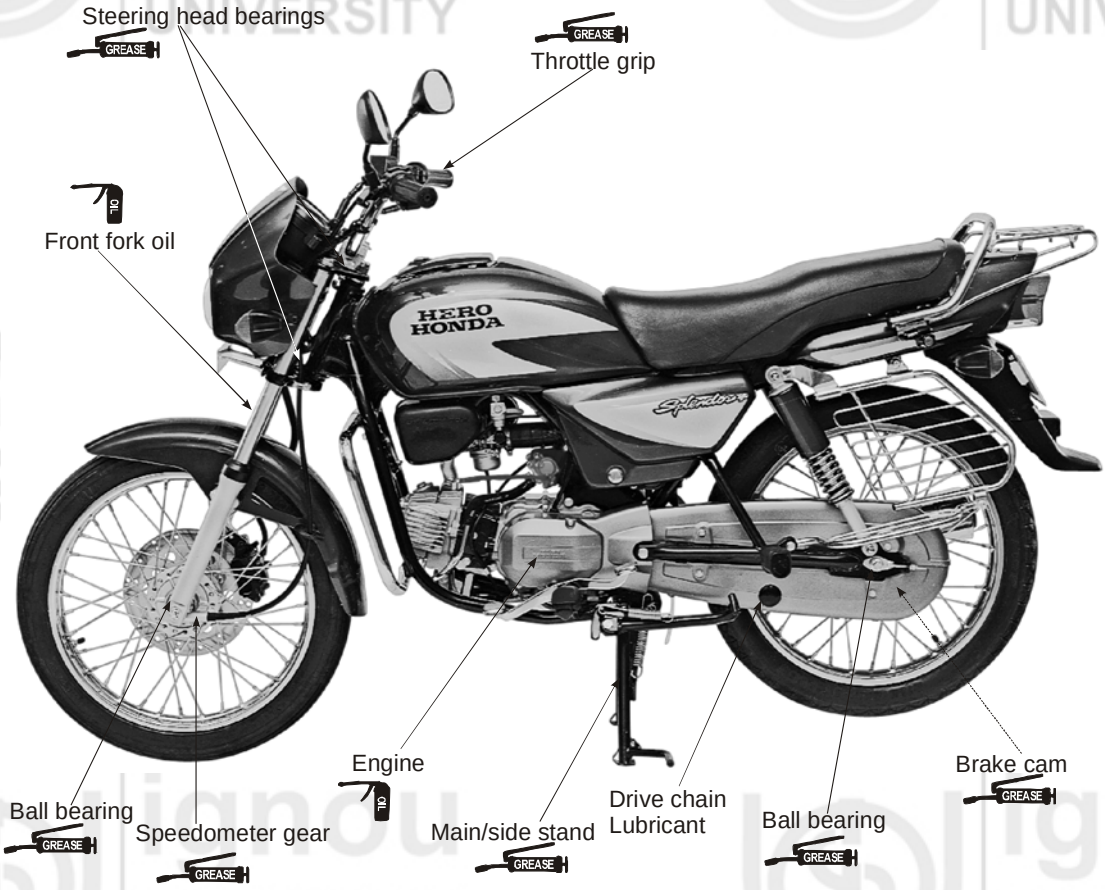


Figure 2.20 : Chassis Lubrication System

Table of Lubrication Points

Sl. No.	Name	Type of Lubricant
1	Ball Bearing	Grease
2	Speedometer Gear	Grease
3	Engine	Engine Oil
4	Main Stand	Grease
5	Brake Pedal	Grease
6	Drive Chain	Oil
7	Front Fork	Oil
8	Throttle grip	Grease
9	Steel Balls, Cone Race,	Grease
10	Ball Race	Grease

SAQ 2

(a) What are the functions of lubrication system?

(b) Define the working of lubrication system used for internal parts of engine.

(c) What is chasis lubrication system? Describe with the help of diagram?

(d) What is a lubricant? List their types and give functions of each of the lubricant types.

(e) What do you mean by following properties of lubricants?

(i) Viscosity, (ii) Viscosity Index, (iii) SAE No.

(iv) SJ Grade

2.2.5 Cooling System

A cooling system is designed to remove a part of heat generated inside the cylinder that may be harmful to engine parts. The heat is produced in the engine cylinder by the burning of fuel. A part of this heat is converted into the useful energy. But the remaining part of heat that cannot be utilised has to be discharged properly from the engine and surrounding parts, otherwise it might be a cause of serious problems in the engine like piston seizure.

A part of unutilised heat escapes with the exhaust gases and lubricants but still some extra heat has to be removed and this part of unutilised heat is removed by the cooling system.

Necessity of Cooling System

- To prevent the engine from excessive heating.
- To avoid the piston seizure due to thermal expansion.
- To prevent burning of lubricant and warping of valves.
- To prevent scoring of cylinders, binding of pistons and bearings.

Types of Cooling System

Two types of cooling system are used in two wheelers.

(a) Natural air-cooling.

(b) Forced air-cooling.

Natural Air Cooling

The strips like extensions (fins) are integrated on the outside of the cylinder and cylinder block. These fins are designed in such a way that they can transfer maximum heat. The fins remain in direct contact with atmospheric air. The air blows over the fins and carries away the heat. When the vehicle travels, the relative speed of air increases, hence the rate of heat removed also increases. In motorcycles and scooters and in several small size engines, natural air cooling is preferred due to its simplicity.

Forced Air Cooling

A fan makes a difference between natural air cooling and forced air-cooling. The fan supplies the required air to carry away heat from the engine cylinder. As indicated by the name of cooling system, the air is forced towards the engine cylinder by the fan that passes over the engine and takes away heat and discharges to the atmosphere. Generally, this type of cooling is used in scooters. In bigger size engines, forced air cooling mechanism or liquid cooling (water or other coolants) system may be required.

SAQ 3

(a) How is the cooling system useful in two wheelers?

(b) Which cooling system is mainly used in two wheelers? Describe working of that cooling system.

(c) Describe the working of an engine.

2.2.6 Transmission System

In a two-wheeler, the transmission system means the whole arrangement that transmits the power from the engine crankshaft to the rear wheel.

Necessity of Transmission System

- The main purpose of Transmission system is to provide a means to vary the torque ratio between the engine and the road wheels. It means the wheel may not be able to run at engine speed every time. So the transmission system provides the required speed.
- The transmission also provides a neutral position, so that the engine and the road wheels are disconnected even with the clutch in the engaged position, a position of idle run of engine.

Concept of Torque (Speed) Variation in Transmission System

When two different size gears are running at same speed, then the small gear makes more revolution than the big gear. This means the speed ratio can vary with the gear size.

The same principle is applied in the speed variation of motorcycle. When smaller gear on the main shaft is in mesh with larger gear on the counter shaft, the revolutions of the counter shaft will be lesser than the main shaft, because the larger gear mounted on countershaft will make less revolution than the smaller gear mounted on main shaft.

The transmission system mainly consists of clutch, gears and drive chain. As shown in Figure 2.21 (diagrammatic representation of a transmission system), a clutch is mounted on the main shaft of the gear box. A primary drive gear mounted on main shaft of the engine drives the driving shaft of the clutch through primary driver gear while the driver shaft of the clutch is connected to the main shaft of the gear box. The drive chain connects the counter shaft and rear wheel. The power is transferred from the main shaft to the counter shaft through various gears and then to the rear wheel through a drive chain.

A higher torque is required to bring the motorcycle into running condition from stationary state but when the motorcycle is running at high speeds the torque required is less. Transmission system is shown in Figure 2.21.

Figure 2.21 : Transmission System

Various Components of Transmission System

- **Primary Drive Gear :** It is mounted on crankshaft. It transfers power to primary driven gear.
- **Primary Driven Gear :** It is an integral part of the clutch outer body. It is mounted on the driving shaft of the clutch.
- **Clutch :** The driven plate of the clutch is mounted on the gear box main shaft. The clutch connects or disconnects the engine crank shaft with the main shaft of the gear box.
- The gears are mounted on both the main and counter shafts of the gear box.
- The Drive Chain finally transmits power to rear wheel from counter shaft of the gear box.

Clutch

Clutch is a part which is used to transmit the rotary motion of one shaft to another shaft, when desired; the axis of second shaft must be in line with the axis of the first shaft (coaxial). The clutch in the motorcycle works on the principle of friction. When a rotating disc comes in contact with other disc, then by the virtue of friction of these plates, both the discs rotate as a single unit.

The clutch may be wet or dry. A wet clutch runs with oil but a dry clutch is not lubricated. A wet clutch will run cool. Further, the clutches may have flat or conical contact surfaces.

In a single plate clutch, contact may be made on two sides while in case of multiple clutches several plates make contact. The other types are semi-centrifugal and centrifugal clutches.

Multi plate clutches are used in motorcycles since they require less radial space.

Various Factors in Clutch Design

- **Torque Transmission :** The clutch should be able to transmit the maximum torque of the engine under all conditions. It is usually designed to transmit 125 to 150 percent of the maximum engine torque. As the clutch slips during engagement, the clutch facing is heated. Clutch temperature is the major factor limiting the capacity of a clutch. This requires that the clutch facing must maintain a reasonable coefficient of friction with the mating surfaces under all working conditions. Moreover, the friction material should not crush at high temperatures and clamping loads.
- **Gradual Engagement :** The clutch should positively take the drive gradually without the occurrence of sudden jerks.
- **Heat Dissipation :** During clutch application, large amount of heat is generated. The rubbing surfaces should have sufficient area and mass to absorb the generated heat. The proper design of the clutch should ensure proper ventilation or cooling for adequate dissipation of heat.
- **Vibration Damping :** Suitable mechanism should be incorporated within the clutch to eliminate (to reduce it drastically) noise produced in the transmission.
- **Size :** The size of the clutch must be smallest possible so that it should occupy minimum amount of space.
- **Inertia :** The clutch rotating part should have minimum inertia. Otherwise, when the clutch is released for gear changing, the clutch plate will keep on spinning, causing hard shifting and gear clashing in spite of synchroniser (the same speed).
- **Clutch-free Pedal Play :** To reduce effective clamping load on the carbon thrust bearing and wear thereof, sufficient clutch free pedal play must be provided in the clutch.
- **Ease of Operation :** For higher power transmissions, the operation of disengaging the clutch must not be tiresome to the driver.

Working of Clutch

The clutches are operated mechanically through linkages. When operator presses the clutch lever, the pressure on the clutch plates reduces to zero, hence the contact breaks between clutch plates and friction plates and as a result each plate becomes free to move independently and engine is disconnected from gearbox. This helps in smooth gear shifting.

Gear Boxes

Motorcycle engines do not produce their full power at all RPM. There is very little power at idle and the power goes up as the engine RPM increases. At a certain RPM, which is different for each engine, the power goes up quickly and continues to increase until it hits the maximum RPM for that engine. This is called the “Power Band”. The gear boxes are provided to use this power transition properly. A gear box consists of arrangement of gears and can be classified into three types : (a) Constant mesh type; (b) Synchro mesh type; and (c) Sliding mesh type.

Constant Mesh Type Gear Box

Most of the gearboxes on motorcycles are “constant mesh”. That means all the gears are constantly meshed with one another and are always spinning. The four/five gear constant mesh type gearbox is employed in Hero Honda Motorcycles.

In the motorcycle gearbox, there are two shafts, the one with the clutch on its one of the ends is the main shaft and the other is the countershaft and it has the countershaft sprocket on one of its ends.

Some of the gears are fixed and/or machined on the transmission shafts. The other gears slide or spin on these shafts. The gears that slide back and forth on the shafts are called Dogs. These Dogs are knobs of metal that fit into holes in the gears beside them, locking the two gears together. The sliding gears are moved back and forth by Shifter Forks which in turn are moved by a shifter drum. The Shifter Forks are usually attached to the Shifter Drum or move in a groove machined in the Shifter Drum. This drum is turned or rotated through different styles of linkages, by the gear Shifter Shaft. As the gears move back and forth they lock the main shaft and counter shaft together through each set of gears, giving us different gear combinations, e.g. 1st, 2nd, 3rd, etc.

Gears

A gear is a disc with teeth on its boundary. If these teeth are formed at its inner boundary then this gear is known as an internal gear (annulus). These types of gears are used to drive other gears in the same direction. A gear that has teeth on its outer boundary periphery is known as external gear and drives mating gear in opposite direction.

Shifting of Gears

To obtain different size combinations of gears, the gears slide on their shafts to mesh with different gears. The position of different gears, e.g. 1st, 2nd, 3rd, etc. for Hero Honda motorcycle are shown in Figures 2.22-2.26.

Figure 2.22 : Gear Drive Position – (Neutral)

Neutral Position

In neutral position, the fixed gears M1 and M3 move (rotate) free gears C1 and C3, respectively but since, we know that gears C1 and C3 are free hence, they cannot rotate the counter shaft, hence, no power is transmitted to wheels.

First Gear Position

When gear M1 is in mesh with gear C1 then this is first gear drive position. The gear C1 is freely attached to counter shaft, so gear C1 itself cannot rotate the countershaft. It rotates countershaft by the help of fixed gear C2. The fixed gear C2 slides on the countershaft with the help of shifter fork and comes in contact with C1 as a result gear C2 is rotated by gear C1. Since gear C2 is a fixed gear to countershaft, hence it will rotate the countershaft. The small size gear (M1) at main shaft is meshed with large size gear (C1) on countershaft. The bigger gear (C1) will make less number of revolutions than smaller gear (M1), hence the countershaft rotates at lower speed than main shaft; therefore the speed of rear wheel will be less than the speed of the engine.

Figure 2.23 : Gear Drive Position – 1st

Second Gear Position

When gear M2 on the main shaft is in mesh with gear C2 on the countershaft then this is known as second gear position. To rotate free gear M2 with the main shaft, it needs a help of fixed gear. The fixed gear M3 slides on the main shaft by the help of shifter fork and comes in contact with the gear M2 and turns the gear M2. Since gear M2 is in mesh with fixed gear C2, hence C2 is turned causing countershaft to rotate.

Third Gear Position

When gear M3 on the main shaft is in mesh with gear C3 on the counter shaft then this is known as third gear position. The fixed gear M3 rotates with main shaft and it rotates free gear C3 but gear C3 cannot rotate countershaft, so fix gear C2 comes in contact with gear C3 and rotates countershaft.

Fourth Gear Position

When gear M4 on the main shaft is in mesh with gear C4 on the countershaft then this is known as fourth gear position. The free gear M4 rotates with the main shaft by bringing M3 in its contact by sliding M3. Thus, C4 will turn causing countershaft to rotate.

Figure 2.24 : Gear Drive Position – 2nd

Figure 2.25 : Gear Drive Position – 3rd

Figure 2.26 : Gear Drive Position – 4th

Means of Starting

Kick Starting

A kick lever is used to crank the engine. When we crank the engine, the air-fuel mixer is sucked into the engine and pressure builds inside the cylinder and at the same time, a spark is ignited that leads to combustion in the cylinder. This process makes the engine to run.

Self-Starting

In a self-starting engine, a motor is used to crank an engine and human effort in cranking is minimal. The motor is powered by a battery.

SAQ 4

(a) What is transmission system? Represent the main components of transmission system with the help of diagram.

(b) What do you mean by terms torque, power and RPM?

(c) What are the types of clutch? State their functions also.

(d) What is a gear box and what are the types of gear boxes?

(e) Describe mechanism of constant mesh type gear box.

(f) What do you mean by a gear?

2.2.7 Fuel Feed System

Fuel feed system is provided to serve the air fuel mixture to the engine in an appropriate ratio so that it can burn efficiently and transmit power to crank shaft. Carburettor is used in the motorcycle to perform some important functions. Carburettor is the most important part of fuel system. The function of carburettor is to supply combustible mixture of fuel and air in correct proportion during all conditions of engine working.

Carburettor

It supplies the air fuel mixture at varying proportions that suits the changing operating conditions of motorcycle. The carburettors for 100 cc and 125 cc motorcycle are respectively shown in Figures 2.27 and 2.28.

Carburettor has following main functions :

- It breaks the fuel into small particles (fine mist). This function is known as atomization. The atomized particles of liquid evaporate to produce vapour of fuel.
- To mix both air and fuel.
- Provide adequate quantity of air fuel mixture according to temperature, load and speed conditions.

Figure 2.27 : Exploded View of a Carburettor for 100 cc Motorcycle

Figure 2.28 : Exploded View of Carburettor for 125 cc Motorcycle

Main Components of a Carburettor

Venturi Tube

A venturi is basically an air passage that has a shape similar to that shown in Figure 2.29 that is narrower at the middle known as venturi tube. As air travels through the narrow part of venturi, its velocity increases and the pressure drops.

Figure 2.29 : Venturi Tube

Throttle Valve

The throttle valves control the quantity of mixture supplied to the cylinder. This is controlled by throttle grip with the help of a cable. When we open throttle valve, the quantity of mixture supplied to cylinder is increased, as a result power generated inside the cylinder is also increased that may increase the vehicle speed.

Choke

This is incorporated in the main air passage and used to control the air quantity to supply rich mixture. While starting an engine, the choke shuts off the air supply to the carburetor so the resulting mixture will be rich in fuel and this opens automatically as the engine heats up.

The ideal carburettor passes a complete vaporized fuel air mixture in the proper proportions into the cylinder. A part of atomized fuel is vaporized in heated spots in the intake manifold.

Working Principle of Carburettor

All carburettors work on one basic principle, the venturi. A venturi is a very simple device that changes air pressure by changing the velocity of the air. As velocity increases, pressure drops and as velocity decreases, pressure rises. As air travels through the narrow part of the venturi, its velocity increases and the pressure drops. If there is a fuel passage at that low pressure point and the fuel reservoir is at a higher pressure, fuel will be pushed into the venturi, i.e. a carburettor needs a pressure difference to allow flow of fuel. Venturi, thus, creates the pressure difference.

We will discuss working of piston type carburettor at various operating conditions that is used in Hero Honda motorcycles.

Idle Circuit

During Idling while the throttle valve is almost closed the vacuum is very high at Venturi area and no air-fuel mixture is drawn through the main circuit, as the jet needle is not lifted from the seat. At idle, the throttle valve is pushed most of the way down by a spring, almost completely closing off the main bore. To keep the idle conditions stable, a slow speed circuit is devised and controlled through the air-screw where the air-fuel mixture is regulated. Because of suction, the air-fuel mixture is delivered in atomized form through the slow jet opening.

Figure 2.30 : Carburettor in Idle Condition

Ideal Speed Circuit (*36 to 40 kmph*)

Motorcycle is brought into this condition by raising the throttle valve and an incorporated jet needle. The air enters through the main passage shown in Figure 2.31 and the fuel enters through the main jet.

Figure 2.31 : Carburettor in Ideal or Normal Condition

High Speed Circuit

The speed of motorcycle is further increased by raising the throttle valve. In this condition, fuel and air enter through main passage and main jet as can be seen in Figure 2.32.

Figure 2.32 : Carburettor in High Speed Condition

Mixture Requirement

Rich Mixture

It generally ranges from approximately 7 : 1 to 13 : 1 proportion of air to fuel by weight. To get a high power rich mixture is required. In fact to start the engine from cold, a rich mixture is required, because fuel does not vaporize completely.

Lean Mixture

It generally ranges from approximately 15 : 1 to 17 : 1. By the use of this fuel, engine runs for higher economy.

SAQ 5

(a) What is the function of fuel feed system?

(b) What are the functions of a carburettor?

(c) List main components of a carburettor.

(d) Describe the working principle of carburettor for the following conditions :

(i) Slow Speed

(ii) Idle Speed

(iii) High Speed

2.2.8 Electrical System

Electrical system consists of a battery, spark plug, lighting system, ignition system, charging system and electricity generator. These components of electrical system are used to fulfill particular functions. This system facilitates a required current for lighting, horn, ignition of fuel in the engine and other purposes.

Necessity of Electrical System

- Lights and horn for safe riding of vehicle even at night.
- Electricity for ignition of fuel in the engine.
- Battery Charging.

Fundamentals of Electricity

Current

The flow rate of charge or electrons with respect to time is known as current. It is measured in ampere (A).

Resistor

The wire that opposes (resists) the flow of current through it by virtue of its property is called resistor. This property is known as resistance. The resistance is measured in ohms.

$$R = \rho L/A$$

where

R = Resistance of wire,

A = cross section area of wire,

L = length of wire, and

ρ = resistivity of wire material.



Resistor

Thus, by increasing length of wire, the resistance increases and by increasing thickness of wire, the resistance decreases.

Voltage

The voltage is the potential difference between two points. The current flows from higher potential to lower potential. Voltage is measured in volts (V).

$$V = IR$$

Alternating Current (*AC*)

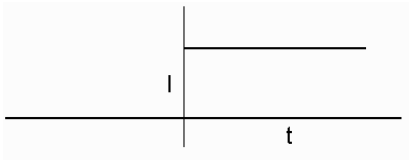
The electricity whose direction changes at regular interval is known as alternating current.



Alternating Current (AC)

Direct Current (*DC*)

The direct current flows through a wire in one direction and it remains same over time. The current from battery is always direct current.



Direct Current (DC)

Diode

It allows the current to flow in one direction only and offers infinite resistance in other direction. It rectifies the alternating current that is why we call it rectifier also.



Diodes

Condenser (*Capacitor*)

A capacitor is used to store the electrical energy. The maximum value of stored energy depends on its capacity. It helps to achieve required high voltage for the sparking.



Condenser

Coil

A coil is used in ignition system to increase the voltage supplied by the battery or magneto.



Coil

Fuse

Fuse wire is made of soft metal and housed in a glass tube. It is connected in circuit. It protects the electrical accessories, which may be damaged by the overflow of current. Whenever there is an overflow of current in the circuit, fuse wire heats up and melts as a result circuit opens at that position and disconnects the component from supply.



Fuse

Zenner Diode

Zenner diode is generally used as a voltage regulator. When voltage falls below the predetermined value, zenner diode functions in the same way as an ordinary diode.



Zenner Diode

Thyristor

The thyristor is just like an ordinary diode.



Thyristor

Ignition System

The ignition system is a source and a means of initial spark required to start burning of fuel. A very high voltage supplied at the spark plug tip causes this spark. It consists of magneto, CDI unit, ignition coil and spark plug. Following are the points of necessity of ignition system :

- Means to provide spark for the running of engine.
- To minimize the arcing between points.

Working Principle of Ignition System

An alternating current (AC) is generated when a magnet moves between two stationary coils. In a magneto, a coil of wires moves between two magnets to produce AC. This AC supply is rectified before use. After rectification this current is also used for other electrical purposes.

The CDI unit consists of a capacitor which is charged through the AC current. When reluctor comes in front of the pulser coil a small amount of current (pulse) is sent to the CDI unit which discharges the capacitor. The discharged current flows through the primary windings of the ignition coil. This induces current in the secondary coil and the voltage is stepped up. This high voltage current jumps across the gap between spark plug electrodes and generates spark.

Spark Plug

A spark jumps between the electrodes of the spark plug when the voltage developed by the ignition system reaches the required value. This electrical spark occurs at a particular piston position and ignites compressed fuel-air mixture in the petrol engine cylinder. The spark plug is one of the most stressed parts on the engine. It is exposed to :

- temperatures ranging from 60°C to 2500°C in the combustion chamber, fluctuating constantly according to the engine operation,
- combustion chamber pressure fluctuates from a partial vacuum (during intake stroke) to 60 kg/cm^{2} (850 psi) (during combustion),
- electrical loading as high as 30,000 volts., and
- corrosive action of the combustion gases.

More often than not, engine defects are attributed to spark plugs. However, in most cases, the spark plugs are not the cause of trouble but they may lie in the malfunctioning of other engine parts/accessories and adverse operating conditions.

Description about Various Sections and Terminologies of Spark Plug

Terminal Nut

This is a detachable type of terminal nut. The high tension (HT) cord is connected here. This is also available in non-detachable type.

Figure 2.33 : Spark Plug

Insulator

This is the top portion of spark plug made of high purity aluminum oxide having excellent characteristics such as good heat conductivity, mechanical strength.

Stud

This is fixed to the insulator. The high-tension cord is also connected to this.

Corrugations

These provide protection against flash over.

Caulked Portion

The insulator and metal shell are caulked together to ensure a gas tight seal.

Metal Shell

It provides a contact for the socket wrench when screwed in.

Plate Finish

It protects against the chemical corrosion.

Gasket

It prevents leakage of the combustion gas. Sufficient tightening torque must be applied.

Sealing Power

Reinforces mechanical strength to prevent leakage under high pressure and temperature.

Battery

Battery is a device that produces direct current (DC). The chemical energy of battery ingredients converts into electrical energy in the form of direct current. This is achieved through some chemical reactions between battery ingredients.

Necessity of Battery

- To obtain power for ignition system.
- To obtain power for lighting.

- To obtain power for horn.
- To obtain power for self-starting.

Lead Acid Battery

The lead acid type of battery is generally used in all types of two wheelers. It consists of six cells and the rating of the battery is 12 V-2.5 Ah and 12 V-5 Ah.

Battery Components

Plates

These are also known as electrodes. There are two types of group of plates – Positive and Negative. Each group contains more than one plate. A group of positive plates is connected to positive terminal and group of negative plates is connected to negative terminal.

Separator

Separator works as insulator between positive and negative plates. These separators are placed between each pair of plates.

Cell Covers

Each cell is covered from the top of the cell covers and the terminals projected from the cell covers. The terminals are arranged alternatively in a line.

Figure 2.34 : Lead Acid Battery

Cell Connectors

A cell connector connects the nearby cell with two terminals of opposite polarity (sign). Post strap connect the plates to the terminal.

Electrolyte

This is the solution of sulphuric acid and distilled water of specified gravity.

Container

It is a hard rubber container divided into compartments with partitions for different cells.

Filler Caps

A filler cap with an air vent is provided for the escape of gases.

Lighting System

The motorcycle lighting system consists of head light, turn indicators, brake lights and panel board lights. Figure 2.35 shows a DC circuit of the electrical system of the motorcycle.

Figure 2.35 : DC Circuit of Motorcycle

SAQ 6

(a) What do you mean by ignition system? Write down its functions.

(b) Briefly describe the working principle of ignition system.

(c) How is energy generated inside the battery?

(d) Describe the construction of lead Acid battery with the help of a diagram.

SAQ 7

(a) What is the function of a spark plug?

(b) What are the types of spark plugs? Point out the differences between them.

(c) Describe the construction of a spark plug with the help of diagram.

(d) Give the use of diode, fuse, capacitor, and thyristor.

2.2.9 Suspension System, Wheels and Tyres

There are numerous types of suspensions for connecting the wheels to frame body. The suspension plays the vital role in handling and driving motorcycle comfortably. People tend to overlook the basics of the suspension.

In addition to connecting the wheel to the frame, the suspension also absorbs shocks from the road surface. If the wheel is directly fixed to the frame, road shock is transmitted directly to the frame, impairing riding comfort and causing damage to the functional parts. By providing a cushion through shock absorber between the wheel and the frame the

motorcycle is protected against shock damage and riding comfort is maintained. The suspension of a motorcycle basically consists of a coil spring and a damper.

First, let us study the functions of the coil spring. The coil spring is installed between the wheel axle and the frame. When the wheel hits the bump the spring is compressed. The compression of spring absorbs the shock and prevents the transfer of these shocks to the vehicle body. This movement of the coil spring is termed as cushion effect.

After compression, the spring rebounds and pushes the frame upwards. As this spring extends, compression occurs again, thus the frame bounces up and down, making the rider uncomfortable.

The damper is a device which suppresses the spring oscillation without impairing the cushion function of the spring. The damper absorbs the energy of oscillations. The absorption of energy by the damper is called the damping effect.

With the combination of coil spring and damper, that is cushion and damping effect, the suspension absorbs the road shock quickly, ensuring the smooth running of the motorcycle.

Necessity of Suspension System

- Not to transmit road shocks to the motorcycle components.
- To guard the driver from road shocks and to make sure the stability of the vehicle in rolling, while in motion.

Types of Suspension System Used in Two Wheelers

Telescopic Hydraulic Fork

This type of suspension system is used to serve for front part of motorcycle. Figure 2.36 (Front Suspension Fork) shows detailed view of front fork assembly. It consists of main and rebound springs, main and bottom tubes, piston, washer, distance tube. A hydraulic fluid fills the space which resists the motion of the piston thus dampens the oscillations.

Figure 2.36 : Front Suspension Fork

Working of Telescopic Hydraulic Fork

When a road shock exerts on to the suspension system it is transmitted to the spring instead of motorcycle frame and the spring starts to oscillate. The springs are supported on the piston which is not allowed to move much by the hydraulic fluid.

Figure 2.37 : Front Suspension Fork

Because of road shock initially piston moves upward and compresses the spring. The damping oil from chamber A transfers to chamber B through a bigger hole as shown in Figure 2.37. During the reverse movement of the piston the damping oil transfers from chamber B to chamber A but this time through a smaller hole. The transfer of oil offers resistance to spring oscillations hence some energy is lost to overcome this resistance. As a result the spring oscillation amplitude (movement) will be lower than before. In this way finally the oscillations of spring will be stopped.

Swing Arm Shock Absorber

This type of suspension is used to serve for rear part of motorcycle. The advantage of this type of suspension is that large deflections are absorbed. It mainly consists of shock absorber, spring at cushion assembly that swings up and down about a pivot when shock exerts on it. The cushion assembly works on the same principle as front fork suspension (Figure 2.38).

Figure 2.38 : Rear Suspension Fork

Wheels

The wheel, along with the tyre has to take the vehicle load, provide a cushioning effect (to protect against force or shock) and cope with the steering control. The wheel must fulfill the following requisite functions :

- It should be strong enough as well as lightest in weight as far as possible.
- It should be replaceable.
- It should be balanced in both static (stationary conditions) and dynamic (moving conditions).
- Its material should not be susceptible to corrosion.

Tyres

Different bikes respond differently to different tyres, so it is impossible to make a generic recommendation. It mainly consists of outer cover and inside tube. The tyre tube assembly is mounted over the wheel rim. The air inside the tube carries the entire load and protects against the shock and force.

Necessity of Tyres

- To supports the motorcycle load.
- To provide cushion against shocks.
- To allow smooth handling of motorcycle.
- To transmit driving and braking forces to the road.

SAQ 8



(a) What is suspension system? Write down its functions.

(b) What are the types of suspension system? Write the names of suspension system used in front and rear of two wheelers.

(c) Describe the working principle of suspension system which is used in the front of motorcycle.

SAQ 9



(a) What is cushioning effect of a tyre?

(b) What are the desired properties of a wheel?

Tyre Outer Diameter

Refer to Figure 2.39.

Tyre Width

The width of the tyre corresponds to the distance between the most protruding portions on both sides.

Overall Tyre Width

Rectilinear distance between both side, including all patterns and character on tyre sidewalls.

Tyre Height

Half of height obtained by subtracting the rim diameter from the tyre outer diameter.

Tread Width

The width of the tyre tread surface. This corresponds, as a rule, to the distance between the most protruding portions on both sides.

Tread Radius

This is also referred to as the crown R. The radius of curvature is expressed in millimeters.

Rim Width

Rim width suitable for effective tyre performance.

Aspect Ratio

Aspect ratio is, as shown in the Figure 2.39, a percentage ratio (oblate ratio) of tyre height to tyre width.

$$\text{Aspect Ratio} = H/W \times 100\ (\%)$$

Figure 2.39 : Tyre

SAQ 10

(a) Describe the structure of tyres.

(b) How are tyres useful in a motorcycle?

(c) How power transfers from the main shaft to countershaft in third and fourth gear positions?

2.2.10 Braking System

A brake is a component of a motorcycle which is used to apply frictional resistance to a moving vehicle to stop or retard it by absorbing its kinetic energy. The functional difference between a clutch and a brake is that a clutch connects two moving parts of a machine whereas a brake connects a moving part to a stationary part.

Necessity of Brakes

- To stop a vehicle in normal as well as emergency conditions.
- To slow down the speed to turn, pass hurdles etc.

Types of Brake

The brakes used in motorcycles can be classified as follows:

(a) Drum Brake

(b) Disc Brake

Drum Brake

The construction of drum brake is clearly shown in Figure 2.40. In a rotating brake drum stationary brake shoes are attached concentric to the axle hub. A back plate is mounted on the axle casing but in case of front axle, the back plate is bolted to the steering knuckle.

The back plate protects the drum and shoe assembly from mud and dust and provides support for the expander, anchor and brake shoe.

Figure 2.40 : Drum Brake

Working of Drum Type Brake

When we apply brake to the motorcycle that means we are applying force 'F' to the brake shoes. This force pushes the brake shoes towards the drum surface. The brake shoe pushes the brake linings which are high friction components against the drum surface. The contact between brake lining and drum reduces the speed of the vehicle. By the virtue of friction between drum and shoes, the kinetic energy (energy of motion) is converted into heat. If we are continuously applying the force on the brakes of the vehicle, the vehicle will stop finally.

Disc Type Brake

As shown in Figure 2.41 the disc brake consists of a pearlitic grey cast iron disc bolted to the wheel hub and a stationary housing called caliper. Casting in two parts makes the caliper, each part contains a piston.

Figure 2.41 : Disc Brake

In the sliding caliper type of disc brakes, there are two parts of piston between which fluids under pressure is sent which then presses one friction pad directly onto the disc, whereas the other pad is pressed indirectly via the caliper.

Comparison of Disc and Drum type Brakes

Drum Brake	Disc Brake
Drum brakes have cooling problem because the friction occurs at internal surfaces, difficult to remove.	Friction surface is in direct contact of air, hence easy cooling.
Non-uniform wear of brake pads because of curved shape.	Uniform wear of pads because of straight shape.
Loss of efficiency of braking due to expansion.	There is no loss of efficiency.
Higher weights.	Lower weights.
Comparatively low anti-fade characteristics.	Comparatively high anti-fade characteristics.
Comparatively complex design.	Simple designs.
Total friction area is comparatively very high almost four times than disc brake.	Low total friction area.

SAQ 11

(a) What are the functions of brake? What are the types of brakes?

(b) How do brakes stop a moving vehicle?

(c) What is the difference between brake and clutch?

(d) Describe and differentiate the working of Drum type and Disc type brakes.

The technical specifications of some of the models launched by Hero Honda Motors Ltd. are presented in **Appendix A** given at the end of this unit.

2.3 SUMMARY

In this unit, we have learnt about the various parts of a motorcycle and their functions. Technical specifications of various models of motorcycles launched by the HHML were also noted. In the next unit, you will learn about various aspects of entrepreneurship development in automobile sector.

2.4 ANSWERS TO SAQs

Refer the relevant preceding text in the unit or other useful books/manuals available on the topic at workshops to get the answers of the SAQs.

Appendix A

Basic Specifications of Hero Honda Motorcycles

Basic Specifications of CD 100 SS Model

DIMENSIONS	
Overall Length	1960 mm
Overall Width	720 mm
Overall Height	1050 mm
Wheelbase	1225 mm
Ground Clearance	165 mm
WEIGHT	
Kerb Weight	112 kg
Maximum Payload	130 kg
ENGINE	
Type	Air Cooled, Four Stroke, OHC
Displacement	97.2 cc
Cylinder Arrangement	Single Cylinder, inclined at 80 degrees from vertical
Maximum Power	7.2 PS at 8000 rpm
Bore X Stroke	50.0 mm X 49.5 mm
Compression Ratio	8.8 : 1
Cylinder Compression	13.5 $\pm$ 1.5 kg/ cm sq.at 1000 rpm
Carburetor	Side Draft Variable - Venturi, Keihin
Fuel Tank Capacity	10.1 Ltrs
Fuel Tank Reserve	2.8 Ltrs
Valve train	Over Head Camshaft (OHC), Poppet Valve
Valve Clearance	0.05 mm when the engine is cold
Starting	Kick Starter
Idle Speed	1400 $\pm$ 100 rpm
Ignition	Capacitor Discharge Ignition (CDI)
Ignition Timing (F mark)	21° BTDC @ 1400 rpm
Full Advance	34.5^{o} BTDC @ 4000 rpm
LUBRICATION	
Oil Pump Type	Pressure Type Wet Sump Lubrication, Trochoid Type Oil Pump
Oil Filter	Wire Mesh and Centrifugal Filter
Air Filter	Oiled Polyurethane Foam
Engine Oil Capacity	0.9 Litres at disassembly, 0.75 - 0.78 lit at draining
Engine Oil Grade	SAE 20 W 40 OIL
Engine Oil make	4T Plus, Automol and Servo Super

TRANSMISSION	
Clutch	Multiplate Wet Type Clutch
Primary Reduction	3.722 (67/18)
Final Reduction	3.000 (42/14)
Gear Box	4 Speed Constant Mesh
Gear	Ratio
First	3.181 (35/11)
Second	1.705 (29/17)
Third	1.238 (26/21)
Fourth	0.958 (23/24)
CHASIS	
Type	Backbone Type
SUSPENSION	
Front	Telescopic Hydraulic Fork
Rear	Swing arm shock absorber assisted
Caster Angle	26 degrees
Trail Length	80 mm
Front Fork Oil Capacity	162-165 ml
BRAKES	
Front	Drum Type
Rear	Drum Type
Brake Fluid	NA
WHEELS AND TYRES	
Tyre Size (Front)	2.75 X 18 - 4 PR
Tyre Size (Rear)	3.00 X 18 - 6 PR
Tyre Pressure	
Front (solo/pillion)	1.5(22) kg/cm^2/1.5 (22) kg/cm^2
Rear (solo/pillion)	2(28) kg/cm^2/2.25(32) kg/cm^2
ELECTRICALS	
Battery	12 V - 2.5 Ah
Alternator	86 W
Spark Plug	MICO-UR4A, NGK-CR7HSA, CHAMPION PRZ7HC
Spark Plug Gap	0.6 mm - 0.7 mm
Head Lamp	25 W/25 W
Tail Lamp/Stop Lamp	5 W/10 W
Turn Signal Lamp	10 W
Fuse	7 Amp

Basic Specifications of CD-Dawn (CD-Deluxe) Models

DIMENSIONS	
Overall Length	1970 mm
Overall Width	720 mm
Overall Height	1045 mm
Wheelbase	1230 mm
Ground Clearance	159 mm
WEIGHT	
Kerb Weight	107 (108) kg
Maximum Payload	130 kg
ENGINE	
Type	Air Cooled, Four Stroke, OHC
Displacement	97.2 cc
Cylinder Arrangement	Single Cylinder, inclined at 80° from vertical
Maximum Power	7.4 PS at 8000 rpm
Bore $\times$ Stroke	50.0 mm $\times$ 49.5 mm
Compression Ratio	8.8 : 1
Cylinder Compression	13.5 $\pm$ 1.5 kg/cm^2 at 1000 rpm
Carburettor	Side Draft Variable - Venturi, Keihin
Fuel Tank Capacity	10.5 ltrs
Fuel Tank Reserve	1.3 ltrs
Valve train	Over Head Camshaft (OHC), Poppet valve
Valve Clearance	0.05 mm when the engine is cold
Starting	Kick Starter
Idle Speed	1400 $\pm$ 100 rpm
Ignition	Capacitor Discharge Ignition (CDI)
Ignition Timing (F mark)	
Full Advance	21° BTDC @ 1400 rpm
	34.5^{o} BTDC @ 4000 rpm
LUBRICATION	
Oil Pump Type	Pressure Type Wet Sump Lubrication, Trochoid Type Oil Pump
Oil Filter	Wire Mesh and Centrifugal Filter
Air Filter	Oiled Polyurethene Foam
Engine Oil Capacity	0.9 ltrs at disassembly, 0.75 - 0.78 ltrs at draining
Engine Oil Grade	SAE 20 W 40 OIL
Engine Oil make	4T Plus, Automol and Servo Super
BRAKES	
Front	Drum Type
Rear	Drum Type

Brake Fluid	NA
TRANSMISSION	
Clutch	Multiplate Wet Type Clutch
Primary Reduction	3.722 (67/18)
Final Reduction	3.000 (42/14)
Gear Box	4 speed Constant Mesh
Gear	Ratio
First	3.181 (35/11)
Second	1.705 (29/17)
Third	1.238 (26/21)
Fourth	0.958 (23/24)
CHASIS	
Type	Tubular Double cradle Type
SUSPENSION	
Front	Telescopic Hydraulic Fork
Rear	Swing arm shock absorber assisted
Caster Angle	26 degrees
Trail Length	79 mm
Front Fork Oil Capacity	162-165 ml
WHEELS AND TYRES	
Tyre Size (Front)	2.75 × 18 - 4 PR
Tyre Size (Rear)	2.75 × 18 - 6 PR
Tyre Pressure	
Front (solo/pillion)	1.75 (25) kg/cm^2/1.75 (25) kg/cm^2
Rear (solo/pillion)	2 (28) kg/cm^2/2.25 (32) kg/cm^2
ELECTRICALS	
Battery	12 V - 2.5 Ah
Alternator	95 W
Spark Plug	MICO-UR4A, NGK-CR7HSA, CHAMPION PRZ7HC
Spark Plug Gap	0.6 mm - 0.7 mm
Head Lamp	35 W/35 W Halogen (Multi Reflector)
Tail Lamp/Stop Lamp	5 W/10 W
Turn Signal Lamp	10 W
Fuse	7 Amp

Basic Specifications of Splendor Plus and Passion Plus Model

DIMENSIONS	
Overall Length	1970 mm (1980 mm)
Overall Width	720 mm
Overall Height	1045 mm (1060 mm)
Wheelbase	1230 mm (1235 mm)
Ground Clearance	160 mm (160 mm)
WEIGHT	
Kerb Weight	109 kg (116 kg)
Maximum Payload	130 kg.
ENGINE	
Type	Air Cooled, Four Stroke, OHC
Displacement	97.2 cc
Cylinder Arrangement	Single Cylinder inclined at 80° vertical
Maximum Power	7.4 PS at 8000 rpm
Bore X Stroke	50.0 mm X 49.5 mm
Compression Ratio	8.8 : 1
Cylinder Compression	13.5 $\pm$ 1.5 kg/cm^2 at 1000 rpm
Carburettor	Side Draft Variable - Venturi, Keihin
Fuel Tank Capacity	10.5 ltrs (12.8 ltrs)
Fuel Tank Reserve	1.3 ltrs (1.1 ltrs)
Valve train	Over Head Camshaft (OHC), Poppet Valve
Valve Clearance	0.05 mm when the engine is cold
Starting	Kick Starter
Idle Speed	1400 $\pm$ 100 rpm
Ignition	Capacitor Discharge Ignition (CDI)
Ignition Timing (F mark)	10° BTDC @ 1400 rpm
Full Advance	30° BTDC @ 4000 rpm
LUBRICATION	
Oil Pump Type	Pressure Type Wet Sump Lubrication, Trochoid Type Oil Pump
Oil Filter	Wire Mesh and Centrifugal Filter
Air Filter	Oiled Polyurethene Foam
Engine Oil Capacity	0.9 ltr at disassembly, 0.75 - 0.78 lit at draining
Engine Oil Grade	SAE 20 W 40 OIL
Engine Oil make	4T Plus, Automol and Servo Super
BRAKES	
Front	Drum Type
Rear	Drum Type

Brake Fluid	DOT - 3 or DOT - 4
TRANSMISSION	
Clutch	Multiplate Wet Type Clutch
Primary Reduction	3.722 (67/18)
Final Reduction	3.000 (43/14) [3.142 (44/14)
Gear Box	4 speed Constant Mesh
Gear	Ratio
First	3.181 (35/11)
Second	1.705 (29/17)
Third	1.238 (26/21)
Fourth	0.958 (23/24)
CHASIS	
Type	Tubular Double cradle Type
SUSPENSION	
Front	Telescopic Hydraulic Fork
Rear	Swing arm shock absorber assisted (with 5 step adjustment)
Caster Angle	26 degrees
Trail Length	89 mm
Front Fork Oil Capacity	162-165 ml
WHEELS AND TYRES	
Tyre Size (Front)	2.75 × 18 - 4 PR
Tyre Pressure	2.75(3.00) × 18 - 6 PR
Tyre Pressure	
Front (solo/pillion)	1.75 (25) kg/cm^2/1.75 (25) kg/cm^2
Rear (solo/pillion)	2.0/2.25 (28/32) kg/cm^2/2.25 (32) kg/cm^2
ELECTRICALS	
Battery	12 V - 2.5 Ah
Alternator	96 W
Spark Plug	MICO-UR4A, NGK-CR7HSA, CHAMPION PRZ7HC
Spark Plug Gap	0.6 mm - 0.7 mm
Head Lamp	35 W/35 W
Tail Lamp/Stop Lamp	5 W/10 W
Turn Signal Lamp	10 W
Fuse	7 Amp